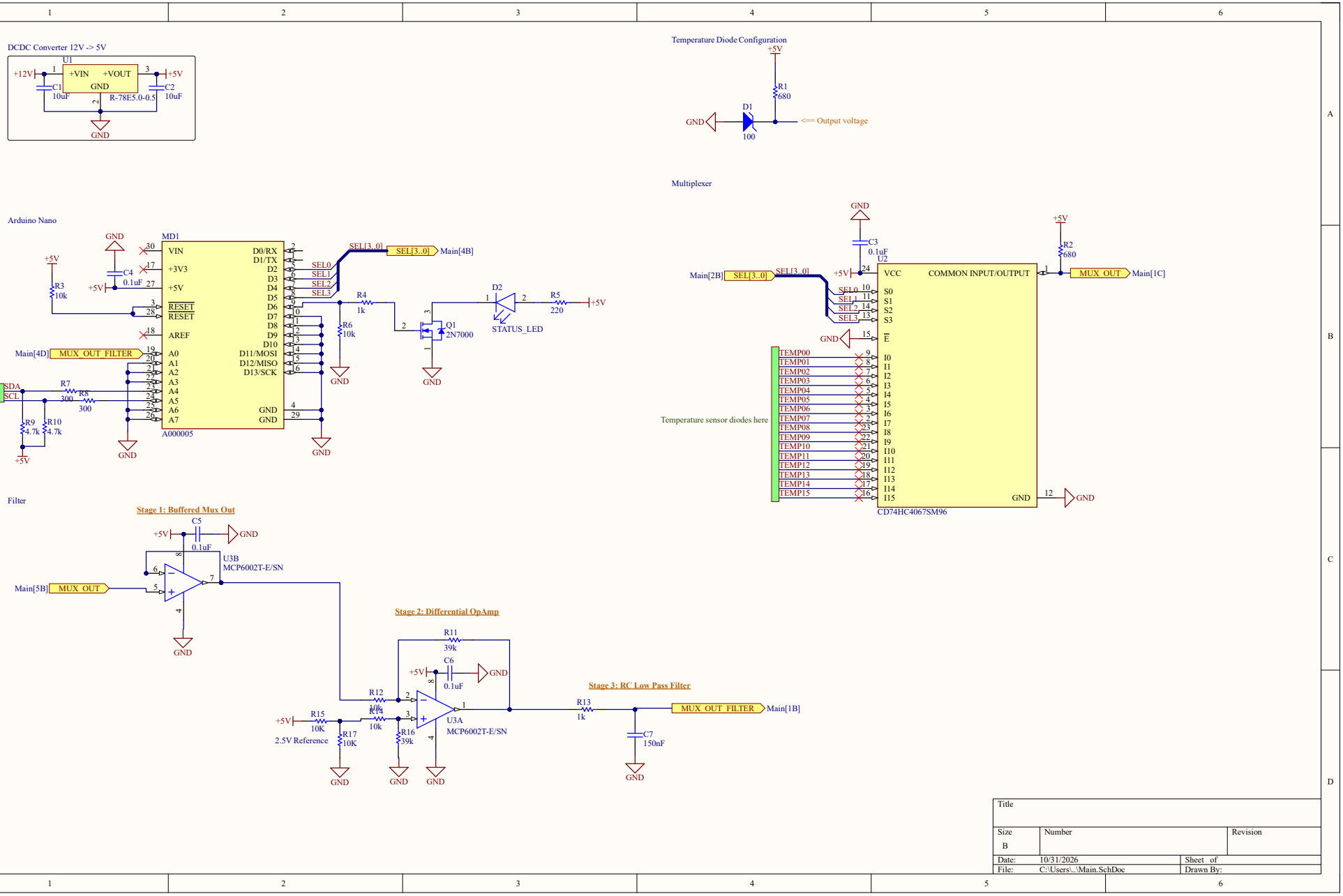

DCDC Converter 12V -> 5V
U1
+12V
+VIN
+VOUT
GND
+5V
C1
10uF
C2
10uF
R-78E5.0-0.5
GND
Temperature Diode Configuration
+5V
R1
680
D1
GND
100
<== Output voltage
Arduino Nano
MD1
GND
+5V
C4
0.1uF
R3
10k
RESET
AREF
VIN
+3V3
+5V
A0
A1
A2
A3
A4
A5
A6
A7
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11/MOSI
D12/MISO
D13/SCK
GND
A000005
SEL[3..0]
SEL0
SEL1
SEL2
SEL3
Main[4B]
R4
1k
R6
10k
Q1
2N7000
D2
STATUS_LED
R5
220
Main[4D]
MUX_OUT_FILTER
I2C Output
SDA
SCL
R7
300
R8
300
R9
4.7k
R10
4.7k
Multiplexer
GND
C3
0.1uF
U2
+5V
R2
680
MUX_OUT
Main[1C]
Main[2B]
SEL[3..0]
VCC
COMMON INPUT/OUTPUT
S0
S1
S2
S3
E
I0
I1
I2
I3
I4
I5
I6
I7
I8
I9
I10
I11
I12
I13
I14
I15
TEMP00
TEMP01
TEMP02
TEMP03
TEMP04
TEMP05
TEMP06
TEMP07
TEMP08
TEMP09
TEMP10
TEMP11
TEMP12
TEMP13
TEMP14
TEMP15
Temperature sensor diodes here
CD74HC4067SM96
Filter
Stage 1: Buffered Mux Out
C5
0.1uF
U3B
MCP6002T-E/SN
Main[5B]
MUX_OUT
Stage 2: Differential OpAmp
R11
39k
C6
0.1uF
R12
10k
R14
10k
R15
10K
R17
10K
R16
39k
U3A
MCP6002T-E/SN
2.5V Reference
Stage 3: RC Low Pass Filter
R13
1k
C7
150nF
MUX_OUT_FILTER
Main[1B]
Title
Size
B
Number
Revision
Date:
10/31/2026
Sheet of
File:
C:\Users\..\Main.SchDoc
Drawn By: